

elections, along with their ideologies, were as follows: PSOE-A, social democracy; PP, liberal conservatism; Cs, liberalism and autonomism; *Por Andalucía* (PorA), left-wing populism and green politics; *Adelante Andalucía* (AA), Andalusian nationalism, left-wing populism, and anti-capitalism; and Vox, right-wing populism, ultranationalism, and national conservatism (Section S2).

In the 2022 elections, the PP won for the first time in all electoral constituencies and also achieved its first absolute majority in the autonomous community. The PSOE-A got its worst result ever in the autonomous community, while Vox failed to fulfil expectations and saw only modest gains. Support for Cs collapsed, with the party being left out of parliament, whereas the left-wing vote divided between PorA and AA platforms ([Wikipedia, 2024d,b](#)).

1.2 Barcelona 2019 City Council Elections

The 2019 Barcelona City Council Election was held on Sunday, 26 May. This election followed the Spanish general elections held a month earlier, which marked the first time Vox secured seats in the Spanish parliament. In the previous City Council elections of 2015 *Barcelona en Comú* (BComú), led by Ada Colau, emerged as the dominant party.

The main candidacies for 2019 elections were: *Esquerra Republicana de Catalunya* (ERC), catalan independence, left-wing nationalism and social democracy; BComú, left-wing populism and participatory democracy; *Partit dels Socialistes de Catalunya* (PSC), socioliberalism; *Barcelona pel Canvi-Ciutadans* (BCNcanvi-Cs), liberal conservatism and Spanish unionism; *Junts per Catalunya* (Junts), catalan independence liberalism; *Partit Popular de Catalunya* (PP), liberal conservatism; *Capgirem BCN-AMUNT* (CUP), catalan independence anti-capitalism and socialism and *Barcelona és Capital-Primàries* (BCAP), catalan independence (Section S3).

Under the leadership of Ernest Maragall, ERC achieved a historic victory in the 2019 elections as it was the first time it had won since the Franco dictatorship. This shift came as the resurgent PSC drew significant support away from the incumbent Ada Colau and her BComú party. Despite this, Colau managed to retain the mayorship by forming an alliance with PSC and securing support from BCNcanvi councillors, led by Manuel Valls. Valls, who had previously run for the French presidency in 2017, was nominated by Cs as their mayoral candidate. Valls's support of Colau's investiture was based on his stated intention to prevent the pro-Catalan independence camp from securing control over Catalonia's capital city ([Wikipedia, 2024c,a](#)).

2 Methods

2.1 Data

The data was collected using academic access to the Twitter API. For Andalusia dataset we use Twitter discussions about the 2022 regional elections held in 19 June. The data includes tweets published between 1 April and 31 July 2022, thus including one month after the elections and two months before them, that includes all the electoral campaign. The tweets included are the ones that contain any of the specific keywords considered, that are general keywords like "19J"; party specific keywords

like “AndalucíaLiberal”, one of the campaign lemmas; as well as the main list heads Twitter accounts of the majority parties that are: @JuanMa_Moreno, @JuanEspadas, @Macarena_Olona, @InmaNietoC, @TeresaRodr_ and @JuanMarin_. In total, there are 8.1 million tweets.

For Barcelona dataset we followed the same collection structure as mentioned before. This includes Twitter discussions about 2019 Barcelona City Council elections held in 26 May recollected from 1 April to 30 June 2019. The tweets included are the ones containing different types of keywords, the ones considered are: general keywords like “Municipals2019Bcn” or “DebatTV3CatRadio”; party specific keywords like “ERCBcn”, “BcnEnComu” or “PSCBarcelona”; and the user account of the heads of the main political parties: @ErnestMaragall, @AdaColau, @JaumeCollboni, @ManuelValls, @QuimForn, @JosepBouVila, @AnnaSaliente, @JordiGraupera and @IGarrigaVaz. In total, there are 3.3 million tweets.

2.2 Communities

We use a community detection algorithm to identify groups of users from the retweets network. Our goal is to obtain communities that describe the political leanings of network users. To do it, we take the largest weakly connected component and use the Clauset-Newman-Moore greedy modularity maximisation (Clauset et al., 2004) to find a community partition. This algorithm begins with each node in its community and repeatedly joins the pair of communities that lead to the largest modularity until no local-scale increase in modularity is possible.

The use of modularity maximisation for community detection has been the subject of significant debate, as highlighted by Peixoto (2021), as it does not fully account for the statistical implications of the optimisation process in identifying the best partition. Nevertheless, it proves valuable in identifying the political alignment of Twitter users based on political party organisations.

2.3 Polarization

Structural polarization assesses the extent to which the network of retweets splits into two distinct and opposing groups. A network is considered polarised if it consists of two communities that have many internal interactions but few interactions between them.

Hence, the same algorithm used for community detection is used to compute polarization but now with a fixed number of clusters of two. We compute the *normalized* (standardised and denoised) polarization to reduce biases related to the network’s characteristic features (Salloum et al., 2022). This polarization is calculated by standardising and denoising the modularity on partitions of random graphs generated with the same configuration as the original network. The value is computed:

$$\hat{\Phi}_z(G) = \frac{\Phi(G) - \langle \Phi(G_{CM}) \rangle}{\sqrt{\langle \Phi(G_{CM})^2 \rangle - \langle \Phi(G_{CM}) \rangle^2}} \quad (1)$$

We compute the change point of the polarization and hate speech percentage time series with the Python package ruptures (Truong et al., 2020). For Andalusia we have

manually set the penalty value to 5 in order to visualise the trend changes due to the election date (June 19, 2022). For Barcelona, the penalty value has been set to 2 for the same reason, which is the election date (May 26, 2019).

2.4 Hate Speech Detection

We define a measure of the amount of hate in the network as the percentage of tweets that contain words that are identified as hate. To compute if a tweet contains hate we use a multilingual BERT model (Aluru et al., 2020) that was trained to classify text inputs as containing hate speech or not with datasets of 5365 and 6000 tweets in Spanish (Basile et al., 2019; Pereira-Kohatsu et al., 2019)¹. This method is subject to several limitations, including the fact that a significant portion of the Barcelona dataset contains tweets in Catalan without the model being trained in this language. However, we know that the algorithm is able to distinguish hate tweets in this language.

3 Results

3.1 Community analysis

After applying the community detection algorithm, we can identify the political alignment of the network groups. This step is done assuming that the retweet is an endorsement of the idea contained in the tweet, and then we name each community with the corresponding political alignment of the accounts it contains. The naming process has been done by analysing the maximum number of user accounts that contain the name of the main political parties competing in the elections. Thus, we named each community with the name of the political party most times represented by the name of the user accounts (Fig. 1).

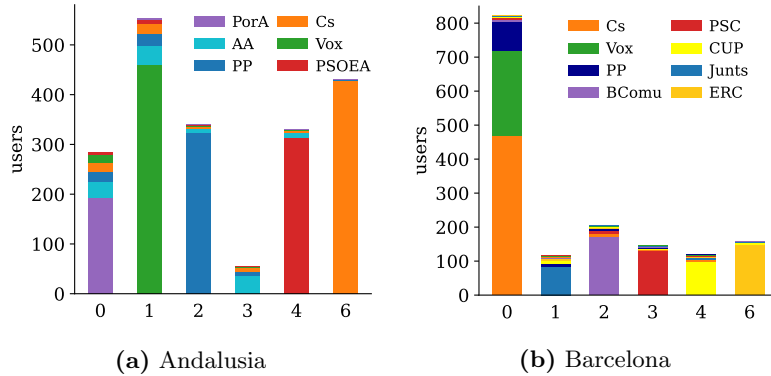


Fig. 1 Number of usernames containing the main political party names for each community.

¹The best validation score achieved by the algorithm trained at different learning rates is 0.740287 for a learning rate of 3×10^{-5} . More details about the model can be found at the official model repository: <https://huggingface.co/Hate-speech-CNERG/dehatebert-mono-spanish>.